Economics 430: Project 1 – Applied Econometric Analysis
Fall 2026, UCLA
Instructor: Dr. Rojas

Due Date: Oct 230 2026

Project Overview and Learning Objectives

This project integrates the econometric concepts covered in the first half of our course with real-world data analysis. By completing this project, you will demonstrate proficiency in Course Learning Outcomes 1-3: following a logical econometric analysis process, using appropriate programming tools, and assessing data characteristics for econometric modeling.

Submission Requirements

The document you submit must consist of a written report including: economic motivation and hypothesis, data description, methodology, results discussion (interpretation of coefficients), limitations, conclusion, references, and Python source code. Submit one report per group with all group members' names included. Upload in PDF or HTML format only. Review the "Assignment Policies" in the course syllabus.

Dataset Selection and Requirements

You may use any dataset of your choice; however, for reference, below are a few examples of publicly available data sources:

- **Housing Market Data:** Median home prices, income, population density, mortgage rates, unemployment (Source: Census Bureau, FRED)
- **Labor Economics:** Wages, education, experience, gender, industry, region (Source: Current Population Survey, Bureau of Labor Statistics)
- **International Trade:** Export/import volumes, GDP, exchange rates, trade agreements, geographic factors (Source: World Bank, OECD)
- **Financial Markets:** Stock returns, market volatility, interest rates, economic indicators (Source: FRED, Yahoo Finance)
- **Public Health Economics:** Healthcare expenditure, income, demographics, policy variables (Source: OECD Health Statistics, CDC)

Dataset Requirements:

- Minimum 500 observations
- At least 5 potential predictor variables
- Cross-sectional data appropriate for OLS analysis

Project Components

1. **Economic Theory and Hypothesis (15 points)**
   (a) State a clear economic research question that your analysis will address
   (b) Provide 2-3 paragraphs reviewing relevant economic theory and citing at least 2 academic sources
   (c) Formulate a specific, testable economic hypothesis about the relationship between your key variables
   (d) Discuss the expected signs and magnitudes of coefficients based on economic theory

2. **Data Description and Exploratory Analysis (25 points)**
   (a) Provide detailed description of your dataset: source, time period, sample selection, variable definitions
   (b) Create comprehensive descriptive statistics table (mean, median, std dev, min, max, skewness, kurtosis)
   (c) Generate appropriate visualizations: histograms, box plots, correlation matrix heatmap, scatter plots of key relationships
   (d) Assess normality using Q-Q plots and density plots for continuous variables
   (e) Identify and discuss potential non-linearities. What transformations might be needed?
   (f) Comment on outliers using appropriate detection methods (IQR, standardized residuals)
   (g) Handle missing values: document patterns, justify imputation method, sensitivity analysis
   (h) Discuss findings from Pandas Profiling analysis

3. **Variable Selection and Model Development (25 points)**
   (a) Apply Boruta Algorithm to identify important predictors (document parameters used)
   (b) Apply Mallows C_p criterion for variable selection
   (c) Check for multicollinearity using Variance Inflation Factors
   (d) Based on economic theory and statistical evidence, develop and justify three competing models:

- Model 1: Simple regression with your theoretically most important predictor
- Model 2: Multiple regression with 2-3 key predictors
- Model 3: Multiple regression including all statistically significant predictors

(e) Compare models using AIC, BIC, adjusted R^2, and cross-validation performance

(f) Select final model and provide detailed economic and statistical justification

4. **Final Model Diagnostics and Robustness (20 points)**

   Required diagnostic tests and visualizations:

   - **Linearity:** Residuals vs. fitted values plots, RESET test for specification
   - **Homoskedasticity:** Breusch-Pagan test, White test, scale-location plots
   - **Normality of residuals:** Q-Q plots, Jarque-Bera test, histograms of residuals
   - **Outliers and influence:** Cook's distance plot, leverage vs. residuals, DFBETAS
   - **Robustness analysis:** Bootstrap coefficient estimates (1000 replications), compare with OLS estimates
   - **Model validation:** 5-fold or 10-fold cross-validation, out-of-sample prediction accuracy

   Document any violations of OLS assumptions and discuss implications for your results.

5. **Results and Economic Interpretation (25 points)**

   (a) Present final model results in professional regression table

   (b) Interpret ALL coefficients in economic terms (magnitude, direction, units)

   (c) Distinguish between statistical significance and economic significance

   (d) Calculate and interpret confidence intervals for key parameters

   (e) Discuss R^2 and model fit in economic context

6. **Limitations, Policy Implications, and Extensions (10 points)**

   (a) Discuss key limitations of your analysis (data, methodology, scope)

   (b) Identify potential sources of bias and how they might affect conclusions

   (c) Describe policy implications of your findings (if applicable)

   (d) Suggest concrete extensions for future research

   (e) Comment on external validity: to what populations/contexts do results generalize?

7. **Code Quality and Reproducibility (10 points)**

   - Well-commented Python code with clear section headers
   - Reproducible analysis (set random seeds where applicable)
   - Efficient, readable code following best practices
   - All figures and tables generated programmatically

Assessment Rubric

Component	**Points**	**Key Criteria**
Economic Theory & Hypothesis	15	Clear research question, literature review, testable hypothesis
Data Description & EDA	25	Comprehensive analysis, appropriate visualizations, missing data handling
Variable Selection & Models	25	Justified model selection, comparison criteria, economic reasoning
Model Diagnostics	20	Complete diagnostic testing, robustness checks, assumption validation
Results & Interpretation	25	Economic significance, coefficient interpretation, literature comparison
Limitations & Extensions	10	Critical assessment, policy relevance, future research directions
Code Quality	10	Documentation, reproducibility, efficiency
Total	**130**	